

Student number: _____

2020 Preliminary Higher School Certificate Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen.
- NESA-approved calculators may be used
- A reference sheet is provided at the back of this paper

For questions in Section II show relevant mathematical reasoning and/or calculations

Total marks -100

Section I- 15 marks (pages 2 - 5)

- Attempt Questions 1 – 15
- Allow about 20 minutes for this section

Section II – 85 marks (pages 6 - 26)

- Attempt Questions 16 – 38
- Allow about 1 hour and 40 minutes for this section

Care has been taken to ensure that this paper is free of errors and that it follows the spirit of the syllabus. Some questions have been adapted from previous Preliminary HSC examinations and various sources, in an attempt to provide students with exposure to a broad range of questions.

Section 1

15 marks

Attempt Questions 1 – 15

Allow about 20 minutes for this section.

Use the multiple – choice sheet for Questions 1 – 15

1. There are six people sharing a house. Each person has a shower each day and uses 125 L of water per shower. Water costs \$3.25 per kilolitre (kL). It is decided to install a water-efficient showerhead that uses 28% less water. How much could be saved on water costs each week?

(A) \$2.10 (B) \$4.78 (C) \$10.75 (D) \$12.29
2. Kim administered a questionnaire to a sample of people by phoning every 100th person in the phone book. What type of sample did Kim use?

(A) categorical
(B) random
(C) stratified
(D) systematic
3. A boat was purchased for \$44 500 and depreciated at 25% of its value each year, as indicated in the table.

Year	Value at the beginning of Year	Depreciation	Value at the end of Year
1	\$44,500	\$11,125	\$33,375
2	\$33,375	\$8,343.75	\$25,031.25
3	\$25,031.25	X	\$18,773.44

The value of X in the table is:

- (A) \$632.81 (B) \$898.44 (C) \$6257.81 (D) \$562.20

4. Calculate the cost of electricity usage where domestic usage is 864KWh at \$0.348500/KWh and off-peak usage is 650 KWh at \$0.140376KWh.

(A) \$392.35 (B) \$401.89 (C) \$506.25 (D) \$670.24

5. Using the table below, find the future value of \$6300 invested at 15% p.a. interest compounding annually for 6 years?

Future Value of of \$1 invested

Periods (years)	Interest rate per period			
	1%	5%	10%	15%
4	1.041	1.216	1.461	1.750
5	1.051	1.276	1.611	2.011
6	1.062	1.340	1.772	2.313
7	1.072	1.407	1.949	2.660

(A) \$5252.80 (B) \$8682.80 (C) \$6894.30 (D) \$14 571.90

- 6 $4.2 \times 10^3 \times 7.1 \times 10^2$ in scientific notation correct to 3 significant figures is.

(A) 2982000 (B) 2.98×10^7 (C) 2.982×10^6 (D) 2.98×10^6

7. A school purchased 300 calculators at \$25 and was given a discount of 15%. The total paid for the calculators was

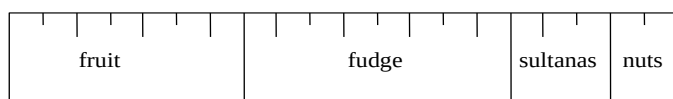
(A) \$1000
(B) \$1125
(C) \$6375
(D) \$7500

8. A section of a wage sheet of a small business shows one employee's details.

Payee	Rate/hour	Normal hours	Overtime (paid time and a half)	Wage
Sam Herd	\$	32	3 hours	\$803.00

The hourly rate of pay that Sam receives is:

- (A) \$ 22.00 (B) \$ 24.00 (C) \$24.50 (D) \$ 26.80
- 9.



The divided bar graph shows the ingredients of a popular chocolate.

What percentage of the chocolate is fudge?

- (A) 10% (B) 15% (C) 35% (D) 40%
10. Joanna delivers advertising pamphlets to houses. The amount P , in dollars, she is paid depends on the number of pamphlets, n , she delivers. Joanna is paid \$5.60 and then \$2.30 for each pamphlet she delivers to a house.

Which one of the following correctly shows the formula for P .

- (A) $P = 5.6 \times 2.3n$
- (B) $P = n(5.6 + 2.3)$
- (C) $P = 2.3 + 5.6n$
- (D) $P = 5.6 + 2.3n$

11. The coordinates of M are:



- (A) $(160^\circ\text{W}, 20^\circ\text{S})$
- (B) $(20^\circ\text{S}, 160^\circ\text{E})$
- (C) $(20^\circ\text{N}, 160^\circ\text{E})$
- (D) $(160^\circ\text{N}, 20^\circ\text{E})$

12. The total cost \$C of a party is given by the formula $C = 65n + 80$, where n is the number of participants.

By how much will the cost increase if the number participating, increased from 55 to 73?

- (A) \$ 275
 - (B) \$970
 - (C) \$1 170
 - (D) \$1 250
13. Each of the 11 letters of the word MATHEMATICS is written on the same size card and placed in a box. A card is drawn and the letter noted. What is the probability that the letter is not a M or A?

- (A) 0
- (B) $\frac{4}{11}$
- (C) $\frac{7}{11}$
- (D) $\frac{8}{11}$

14. Solve the equation $\frac{5x+1}{3} = 7$

- (A) $x = 4$
- (B) $x = 5$
- (C) $x = \frac{16}{3}$
- (D) $x = 12$

15. Brett wishes to invest \$7600 for a period of 7 years. Which of the following would provide the greatest amount?

- (A) Compound interest at 4.4% p.a. compounded quarterly.
- (B) Compound interest at 4.8% p.a. compounded 6-monthly.
- (C) Compound interest at 5% p.a. compounded annually.
- (D) Simple interest at 5% p.a.

Section II

85 marks

Attempt Questions 16-38

Allow about 1 hour and 40 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the end of the paper. If you use this space, clearly indicate which question you are answering.

All necessary working should be shown in every question.

Question 16 (3 marks)

Robert works for two retail shops. He works from 4:00 pm to 9:30 pm on Thursday evenings for 'Cottonworth' and he works from 8:00 am to 10:00 am on a Saturday and 9:00 am to 5:00 pm on Sundays for 'Poles'. The rates paid by each employer are listed below.

Cottonworth	
TIME	RATE
For the first 4 hours / day	\$8.00 / hour
For the next 4 hours / day	\$8.60 / hour
Over 8 hours per day	\$10.60 / hour
Uniform allowance for all employees = \$2.50/ hour	

Poles
Casual rates
Weekdays \$7.50 / hour
Saturday - time and a half Sunday - double time

- (a) What are Robert's weekly earnings from 'Cottonworth'?

1

(b) What are his total earnings for one week?

2

Question 17 (2 marks)



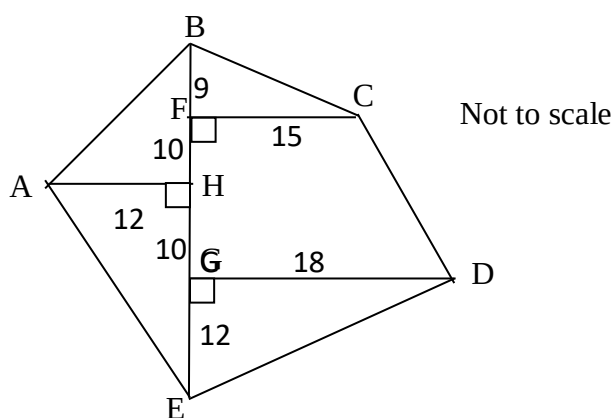
Hemisphere

A child's building block is in the shape of a hemisphere with radius 8 centimetres.

Find the volume of the hemisphere, in cubic centimetres?

Question 18 (4 marks)

A surveyor measured the following offsets on a block of land. All measurements are in metres.



- (a) Find the area of GFCD.

2

- (b) Hence find the area of figure EABCDE.

2

Question 19 (5 marks)

John earns \$56 000 and has allowable tax deductions of \$15 000. Tracey earns \$82 000 and has allowable tax deductions of \$17 000.

Taxable income	Tax payable
\$0-\$6 000	Nil
\$6 001-\$20 000	17c for each \$1 over \$6 000
\$20 001-\$50 000	\$2 380 plus 30c for each \$1 over \$20 000
\$50 001-\$60 000	\$11 380 plus 42c for each \$1 over \$50 000
Over \$60 000	\$15 580 plus 47c for every \$1 over \$60 000

- (a) Find the difference in tax payable between John and Tracey.

3

- (b) If Tracey's employer is deducting \$334.00 per week for tax, will she receive a refund at the end of the financial year or will she have to pay extra tax? Give reasons for your answer. (Assume a 52 week year) 2

Question 20 (4 marks)

Jeff wants to purchase a new boat with a recommended retail price of \$68 000.

- (a) Calculate the stamp duty for the car using the following table. 2

Vehicle value	Rate
\$0 - \$44 999	\$3 for every \$100
\$45 000 and over	\$1350 plus \$5 for every \$100 over \$45 000

- (b) The dealer charges a delivery fee of \$675 on top of a boat registration fee of \$480. Jeff decides to take out comprehensive insurance that costs \$1 260 for the first year. Calculate the total cost of purchasing the boat. 2

Question 21 (3 marks)

The table below shows the domestic electricity prices for a particular town in NSW.

Electricity Tariffs	Rates in cents
Supply charge	66.18 c/day
Usage first 1685 kWh/quarter	24.75 c/kWh
Remaining usage/quarter	27.81 c/kWh

Calculate the amount charged for a quarter (91 days) if the usage is 2352 kWh.

Question 22 (2 marks)

Solve the following equation: $6(2 - 3x) = -60$

Question 23 (4 marks)

For lunch, Cathy decides to visit her local restaurant. She can choose one item from each course of a set menu.

Entree	Main Course
Bread (B)	Steak (S)
Greek Salad (G)	Ravioli (R)
Black Forest Ham (H)	Fish (F)
Salmon Cakes (C)	

- (a) Use a table to list all the different two-course lunches available.

2

- (b) How many two-course lunches are available?

1

- (c) Find the probability that the Greek Salad and Fish combination is chosen.

1

Questions 16-23 are worth 27 marks

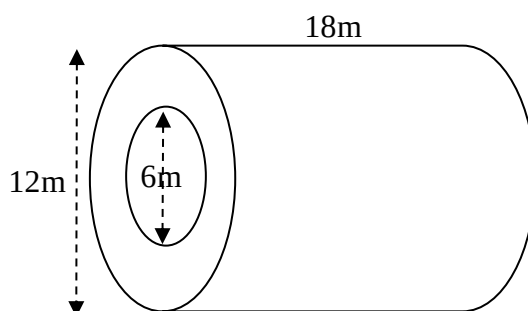
Question 24 (2 marks)

Computer equipment with an original value of \$8 000 depreciates by \$680 every year.

Use the straight line depreciation formula to calculate when the computer equipment is half its original value .

Question 25 (5 marks)

The diagram shows a large concrete pipe with very thick walls.



The entire surface (outside, inside and ends) of the pipe has to be given a coat of paint.

One litre of paint covers 16 square metres. The paint can only be bought in 20 litre cans, each costing \$134.

- (a) Find the total surface area to the nearest m^2 .

4

(b) How much will it cost to paint the concrete pipe?

1

Question 26 (3 marks)

The formula $C = \frac{yA}{y+12}$ calculates a child's dose of medicine (C) in millilitres, where y is the child's age in years and A is the adult dose in millilitres.

(a) Make A the subject of the formula.

2

(b) Find the adult dose of the medicine if a 4-year old child's dose is 12 mL.

1

Question 27 (4 marks)

Trudy is to fly from Melbourne to Vancouver. Melbourne is 10 hours ahead of UTC (Coordinated Universal Time) and Vancouver is 8 hours behind UTC. The plane stops at Hawaii (10 hours behind UTC) for a two hour fuel stop.

The flight from Melbourne to Hawaii takes 10 hours and the flight from Hawaii to Vancouver takes 8 hours.

- (a) What time is it in Hawaii when it is 9 a.m. on a Saturday in Melbourne? **1**

- (b) If Trudy arrives in Vancouver at 10 p.m. on a Sunday, what time did the plane leave Melbourne? **3**

Question 28 (3 marks)

- (a) Paula and Tim were at a party for 7 hours, Paula, who weighs 63 kg had 8 standard drinks and Tim, who weighs 78 kg, had 11 standard drinks.

$$BAC_{male} = \frac{10N - 7.5H}{6.8M}$$

or

$$BAC_{female} = \frac{10N - 7.5H}{5.5M}$$

N is number of standard drinks consumed

H is number of hours of drinking

M is person's mass in kilograms

Using the above **Blood Alcohol Content Estimates** calculate the blood alcohol content (BAC) for both Paula and Tim?

2

- (b) Using the formula Number of hours (N) to reach zero BAC ($N = \frac{BAC}{0.015}$), determine the time taken (in hours and minutes) for Tim to reach zero BAC.

1

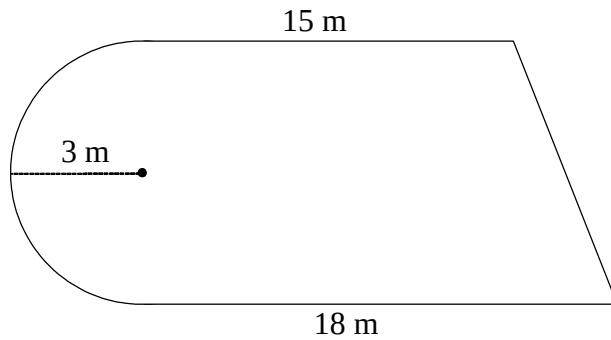
Questions 16-28 are worth 44 marks

Question 29 (3 marks)

The following shape shows the fencing required to surround a park.

- (a) Calculate the length of fencing required, to two decimal places.

2



NOT TO
SCALE

- (b) Fencing costs \$89.65 a metre. What will be the cost of fencing the park area?

1

Question 30 (3 marks)

The approximate braking distance (d metres) can be found using the formula
 $d = 0.00565v^2$, where v km/h is the speed of the car. Find the speed of the car
 if the braking distance is 106.4 m

Question 31 (5 marks)

The teachers at a local high school surveyed their students to see if they
 were using their home computers for school work. The stem and leaf plot show how
 19 students spent their time using their computers on school work at home. Each time
 is recorded in minutes.

Stem	Leaf
0	4
1	2 4 6 8 8 8
2	3 4 4 5 5 5 5 6
3	2
4	2 2 4

- (a) What is the mean time spent doing homework? **1**

- (b) Find a five-number summary for the above data **2**

- (c) Draw a box-plot for the data on homework.

2

Question 32 (5 marks)

A tennis player gets a second serve if the first serve does not go in. Flash Barty, a well-known female tennis player, has a probability of 0.75 of her first serve being successful, and her second serve has a probability of 0.95 of being successful.

- (a) Draw a tree diagram for the first and second serve, showing the probability value on each branch?

2

- (b) What is the probability that at least one of her serves goes in?

Write your answer as a percentage correct to two decimal places.

2

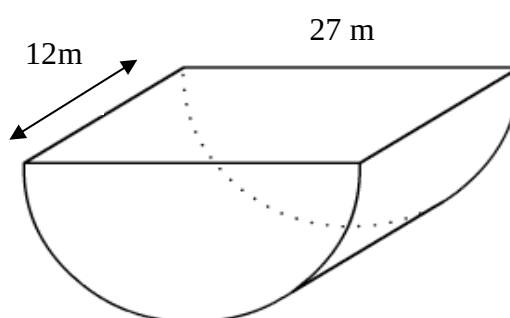
- (c) Find the probability that she serves a double fault. A double fault occurs when both the first and second serve do not go in.

Write your answer as a percentage correct to two decimal places.

1

Question 33 (5 marks)

A cylindrical shape swimming pool is to be constructed as shown below.



- (a) Find the volume of this pool in cubic metres. Give your answer to the nearest cubic metre. 2

- (b) Calculate the capacity of the pool, in kL, when it is completely full. 1

- (c) If the owner wishes to tile the inside surface of this pool, how many square metres will this amount to? (Answer correct to one decimal place) 2

Question 34 (6 marks)

- (a) Complete the cumulative frequency in the following table.

1

Mark	frequency	Cumulative Frequency
10	40	
20	20	
30	80	
40	20	
50	20	

- (b) Construct a cumulative frequency histogram and polygon.

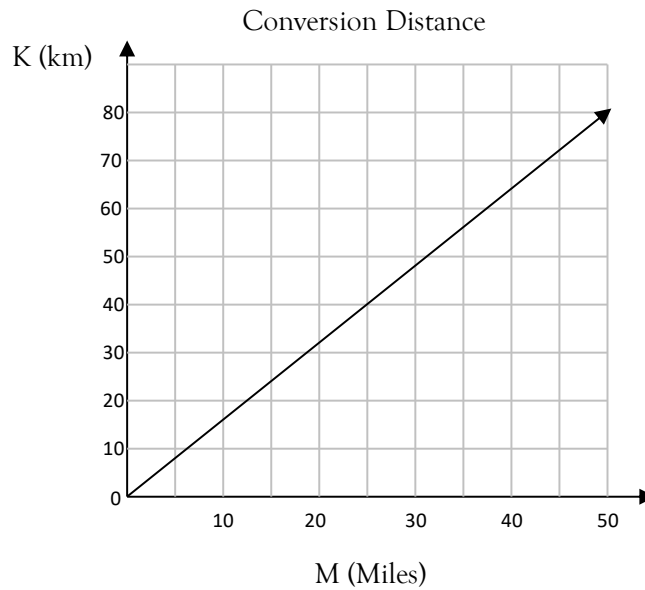
3

- (c) Use the graph to find the median.

2

Question 35 (4 marks)

The conversion graph shown below is used to convert miles to kilometres.



- (a) Find the constant of variation

1

- (b) Write the equation for which K and M are related.

1

- (c) Using your equation convert 100 kilometres to miles.

2

Question 36 (5 marks)

Leila is a sales representative at Zina's Real Estate agency. The agency charges commission on all sales as follows:

Zina's COMMISSION RATES	
Amount	Rate
First \$180 000	8%
Remainder over \$180 000	7%

Leila is paid a retainer of \$560 per week plus 9% of the commission received by Zinas.

In the first week of July 2019, Leila sold one property for \$290 000 and one for \$300 000.

- (a) What is the commission paid to Zinas on the sale of the property worth \$290 000 **1**

- (b) What is the total commission paid to Zinas on the sale of both properties? **2**

- (c) Calculate Leila's pay for the first week in July 2019. **2**

Question 37 (3 marks)

Sarita's car uses 13 litres of petrol to travel 166 *km*. Petrol costs 96 cents per litre.

How far can he drive using \$46.60 dollars of petrol? (Write your answer to the nearest km)

Question 38 (2 marks)

If $S = \frac{n}{2}[2a + (n-1)d]$ change the subject to d

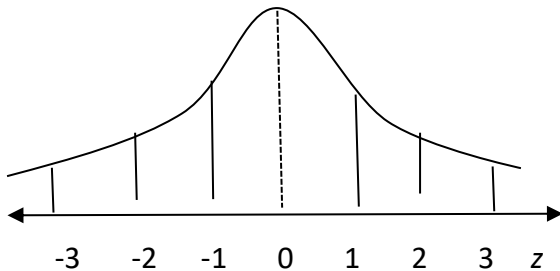
End of Exam

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

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Mathematics Standard 1 and Standard 2 Reference sheet

Measurement Limits of accuracy Absolute error = $\frac{1}{2} \times \text{precision}$ Upper bound = measurement + absolute error Lower bound = measurement – absolute error Length $l = \frac{\theta}{360} \times 2\pi r$ Area $A = \frac{\theta}{360} \pi r^2$ $A = \frac{h}{2} (a + b)$ $A \approx \frac{h}{2} (d_f + d_l)$ Surface area $A = 2\pi r^2 + 2\pi r h$ $A = 4\pi r^2$ Volume $V = \frac{1}{3} Ah$ $V = \frac{4}{3} \pi r^3$ Trigonometry $\sin A = \text{opp/hyp}$, $\cos A = \text{adj/hyp}$, $\tan A = \text{opp/adj}$ $A = \frac{1}{2} ab \sin C$ $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ $c^2 = a^2 + b^2 - 2ab \cos C$ $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$	Financial Mathematics $FV = PV(1 + R)^n$ Straight-line method for depreciation $S = V_0 - Dn$ Declining-balance formula for depreciation $S = V_0(1 - r)^n$ Statistical Analysis An outlier is a score less than $Q_1 - 1.5 \times IQR$ or more than $Q_3 + 1.5 \times IQR$ $z = \frac{x - \bar{x}}{s}$ Normal distribution  - Approximately 68% of scores have z-scores between -1 and 1 - Approximately 95% of scores have z-scores between -2 and 2 - Approximately 99.7% of scores have z-scores between -3 and 3
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